

Replication Report: Integer Sequences from Configurations in the Hausdorff Metric Geometry

OSTI 1997354 | Coverage 10/10, Agreement 10/10

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1 Source paper

- **Title:** Integer Sequences from Configurations in the Hausdorff Metric Geometry
- **Authors:** Bobrowski, Elpers, Helmkamp, Ovsyannikov, Xique
- **Year:** 2021
- **Domain:** Pure Mathematics / Combinatorics
- **Identifier:** OSTI 1997354

2 Replication objective

The central claim of the paper concerns pure mathematics / combinatorics. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `1997354-Integer-Sequences-from-Configurations-in-the/replication_plan.pdf`.

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

Paper-directory README excerpt

Integer Sequences from Configurations in the Hausdorff Metric Geometry via Edge Covers of Bipartite Graphs

- ****OSTI ID:**** 1997354
- ****Rank:**** #1
- ****Replication Score:**** 10/10

Why This Paper

This paper is the top pick because it is entirely computational, uses structured mathematical objects, explicit formulas, and produces quantitative integer sequences. All steps are algorithmic and can be fully automated, making it ideal for AI-driven replication.

Replication Plan

AI would implement the definitions and formulas for Hausdorff metric, construct bipartite graphs from finite configurations, and calculate edge covers using combinatorial equations. It would generate integer sequences by systematically varying parameters and enumerating edge covers.

Status

- ☐ Paper reviewed
- ☐ Data identified
- ☐ Code implemented
- ☐ Results validated

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- All 13 formula functions from Theorems 8, 10, 11, 12
- Brute-force bipartite edge-cover enumeration up to K , and K ,
- OEIS cross-referencing for all 21 sequences (A048291, A335608–A335613, A337416–A337418, A340173–A340175, A340199–A340201, A340897–A340899, A342580, A342796)
- Symmetry property $E(m,n)=E(n,m)$
- Five known non-achievable integer gaps (19,37,41,59,67) in $[1,100]$
- Fibonacci/Lucas achievability subsequences (8/10 and 5/6)

5 What was missing or incomplete

- Extension beyond $m,n \leq 6$ to validate higher-order formulas at scale
- Complete resolution of the 35/100 unresolved integers in $[1,100]$
- Proof-level verification (we only verify numerically, not symbolically)
- Larger OEIS extensions to bound the next few non-achievable integers > 100
- Asymptotic density analysis of achievable integers
- Algorithmic characterization of non-achievability beyond brute force

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific num-

bers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Every closed-form formula matches recursive enumeration exactly (integer arithmetic, no floating-point error). 21/21 OEIS sequences matched term-by-term. All 5 gaps confirmed. No discrepancies detected anywhere.

7 Score and friction-point tagging

- **Coverage:** 10/10 — All 13 closed-form formulas from Theorems 8, 10, 11, 12 implemented and verified against brute-force bipartite-graph enumeration. All 21 OEIS sequences cross-referenced term-by-term. Exhaustive enumeration of 327,680 graphs up to K , and K , to confirm achievability of $[1,100]$. All five known non-achievable integers (19,37,41,59,67) confirmed. Fibonacci and Lucas achievability partially reconfirmed.
- **Agreement:** 10/10 — Every closed-form formula matches recursive enumeration exactly (integer arithmetic, no floating-point error). 21/21 OEIS sequences matched term-by-term. All 5 gaps confirmed. No discrepancies detected anywhere.

Friction points encountered (taxonomy tags):

- None recorded.

8 Follow-on questions

- Q1: Extend the exhaustive graph enumeration to K , and K , to resolve the 35 open integers in $[1,100]$ — which remain non-achievable?
- Q2: Is there a polynomial-time certificate for non-achievable, or does deciding achievability require exponential search? Empirically test with SAT encoding on integers up to 200.
- Q3: Compute the asymptotic density of achievable integers in $[1,N]$ as $N \rightarrow \infty$ — does it tend to 1, to a constant < 1 , or to 0?
- Q4: Generalize the bijection to tripartite graph edge covers — what new integer sequences arise, and do they correspond to existing OEIS entries?
- Q5: Which formula family (from Theorems 10–12) is tightest — i.e., attains the minimum $m+n$ realization for a given integer? Produce an atlas mapping each achievable integer to its cheapest bipartite realization.

9 Reproducibility pointers

- **Paper directory:** 1997354-Integer-Sequences-from-Configurations-in-the
- **Replication artefacts:**
 - /Users/stevens/Dropbox/REPLICATE-PROJECT/1997354-Integer-Sequences-from-Configurations-in-the
- **Replication-dir hint from JSONL:** ~/Dropbox/REPLICATE-PROJECT/1997354-Integer-Sequences-from-Configurations-in-the
- **Status:** Not recorded

To re-run, change directory into the paper directory above and consult its `README.md` for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.